# BOOK II — SCIENTIFIC INSTRUMENT MASTER

**Document ID:** PH6-BOOKII-CONSOLIDATED-1.0

**Role:** CANON

**Status:** ACTIVE

**Question Answered:** How truth is measured.

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## B2 § 1 — Deterministic Instrument Science Objectives and Scope

PH6 is a scientific instrument. Objectives: Preservation, Determinism, Crash safety, Auditability, Scientific use, Authority isolation, Export sovereignty.

## B2 § 2 — Six-Engine Runtime Model (Lane + Layer Architecture)

- **Engine I (PSEUDO):** Deterministic authority engine.
- **Engine II (Runtime):** FAST execution.
- **Engine III (Persistence):** HOT append-only.
- **Engine IV (Topology):** SoSo advisory.
- **Engine V (Swarm):** Runtime blocked.
- **Engine VI (Instrument Science):** PSEUDO-SCI sideband.

## B2 § 3 — Calibration Doctrine

Deterministic metrics must be calibrated to sensor-specific baselines without allowing runtime adaptation.

## B2 § 4 — CRAM Storage Model

- **CRAM-0:** Write-first intake buffer.
- **CRAM-A:** Immutable authoritative evidence (PASS).
- **CRAM-R:** Reject vault (DROP).
- **MRAM-S:** Advisory shadow storage (Lane 2).

## B2 § 5 — Sensor Normalization

All sensor inputs must be normalized to deterministic fixed-point representations before PSEUDO-M processing.

## B2 § 6 — evaluate_frame Contract

Returns: `verdict` (PASS/DROP), `metrics` (fixed-point), `reasons`, `input_hash`, `gate_profile_version`, `threshold_profile_version`.

## B2 § 7 — Entropy / Laplacian / Motion Gates

- **Entropy:** $3.5 \leqslant H \leqslant 7.5$
- **Laplacian Variance:** $Var \geqslant 60.0$
- **Motion Fraction:** $0.05 \leqslant M \leqslant 0.95$

## B2 § 8 — PSEUDO Measurement Doctrine

- **PSEUDO-M:** Measures Shannon Entropy, Laplacian Variance, Motion Fraction.
- **PSEUDO-A:** Adjudicates based on fixed thresholds. Any veto → DROP.

## B2 § 9 — EvidencePacket and Schema Definitions

Sealed boundary object containing: `cram_object_id`, `source_id`, `timestamp_utc`, `seq_num`, `sha256`, `size_bytes`, `cram_tier`, `gate_scores`, `verdict`, `veto_reasons`, `policy_version`, `canon_locked`.

## B2 § 10 — Absolute Constraints

- No AI verdict authority.
- No runtime learning in Lane 1.
- No adaptive threshold drift.

## B2 § 11 — Functional and Non-Functional Requirements

(Consolidated from Operational Source Canon v2.0 implementation sections)

## B2 § 12 — Atomic Write Contract

1. Write raw data to CRAM-0.
2. Compute metrics and verdict.
3. Write EvidencePacket to CRAM-A/R.
4. Update authority chain hash.

## B2 § 13 — Standard Operating Procedures

Procedures for intake, preservation, and export (RSYNC).

## B2 § 14 — Authoritative Science Lane (Lane 1)

The primary path for evidence generation.

## B2 § 15 — Exploratory Science Lane (Lane 2)

The path for hypothesis and advisory analysis.

## B2 § 16 — Recovery and Debugging

Crash-consistency ensures that power loss does not create partial authoritative state.